

Production & Exploration Dashboard

Design and Predictive Analytics Concept

Prepared for	Oil India Limited — leadership review
Date	June 2026
Scope	Production and exploration dashboard pages — KPIs, insights, and predictive scenarios
Data sources	Internal: 10-year production & reserves, drilling & workover, FY25-26 MIS, Annual Reports FY21-FY25. External: oil-india.com investor disclosures and SEBI LODR filings (cited inline).

Note on forecast content. All scenarios and predictive panels in this pack are illustrative; coefficients and analog volumes are placeholders pending model fitting and subject-matter-expert calibration. Forecast regions are clearly demarcated. Booked reserves and KPI computations exclude all scenario / unbooked items by design.

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Appendix — for the development team

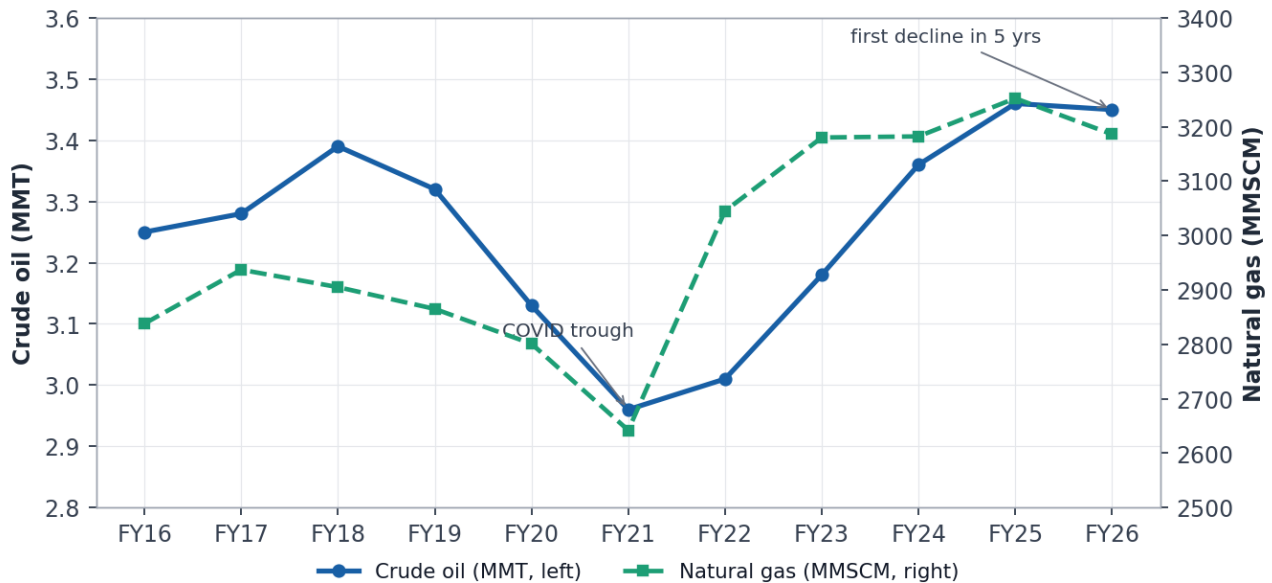
A1	AI extraction discipline — failure modes to prevent
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Visual conventions used throughout:

Source	File, sheet and row references for every figure shown.
Assumption	Stated assumption or inference where a value is derived rather than directly read from a file.
FORECAST	Shaded region on a chart, beyond actuals, indicating illustrative scenario projection.

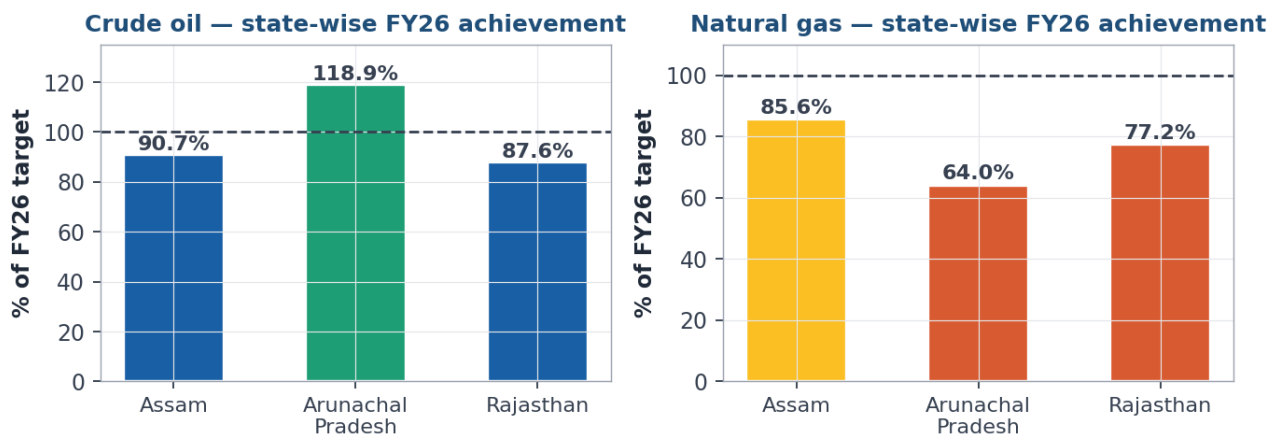
1 Production page — KPIs and ten-year trend

CRUDE OIL 3.45 MMT -0.3% YoY · FY26 · first decline in 5 yrs	NATURAL GAS 3,186 MMSCM -2.0% YoY · FY26	TARGET ACHIEVEMENT 90.9% Crude · 87.1% gas with JV	RESERVE REPLACEMENT 0.94 FY25 · below 1.0 for 3rd year
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Source: 10_Years_Production_and_Reserves_Data.xlsx, Sheet1 rows 3–13 (columns B crude, D gas). FY26 figures from FY202526_Performance.xlsx, sheet Annexure-V-Production, rows 15 (crude) and 28 (gas).

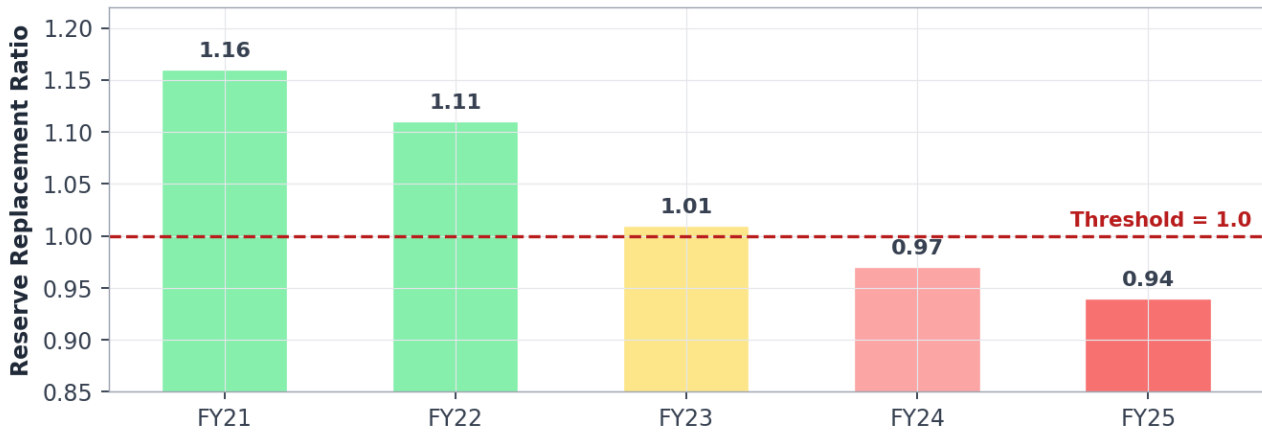
Insight 1. Recovery from the FY21 COVID trough — a 17% rebound through FY25 — has plateaued. FY26 printed the first decline in five years for both crude and natural gas. The state-wise FY26 achievement below shows where the year fell behind: gas at 85.6% of the Assam target is the principal drag.



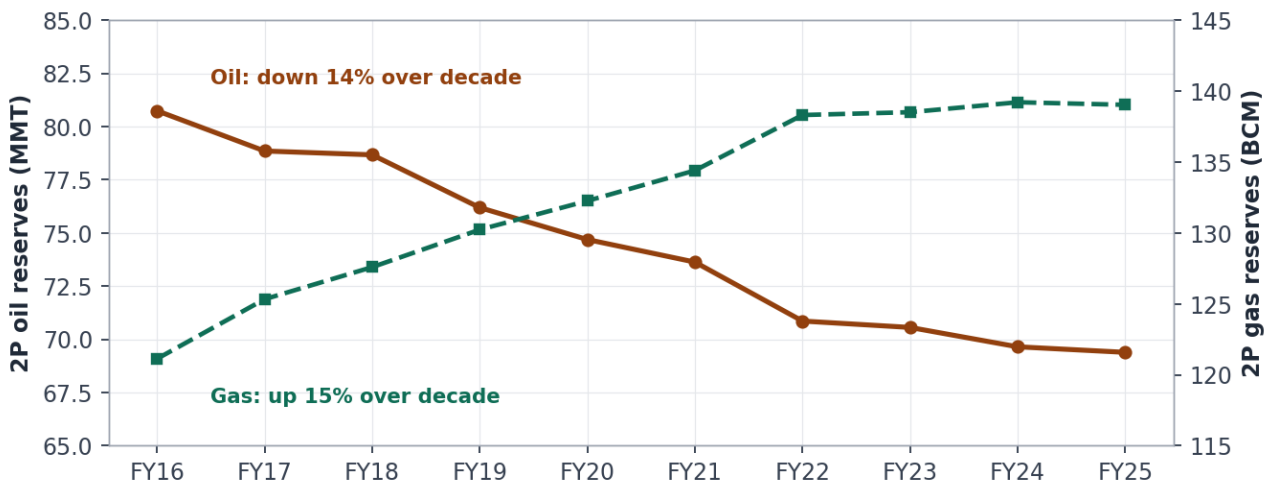
Source: FY202526_Performance.xlsx, sheet Annexure-V-Production: crude state-wise rows 6–8, gas state-wise rows 22–24. Cumulative actuals through March 2026 against annual BE targets.

Assumption: The MIS records FY26 target-versus-actual percentages, not historical decline rates. Any "natural decline" framing for the Assam fields is inferred from the long-running production profile, not stated in the file. State-wise annual production for FY16–FY25 needs extraction from each Annual Report operations section to establish field-level decline curves.

2 Reserves analysis — RRR and 2P reserves divergence



Source: 10_Years_Production_and_Reserves_Data.xlsx, Sheet1 column N "RRR", rows 8–12.



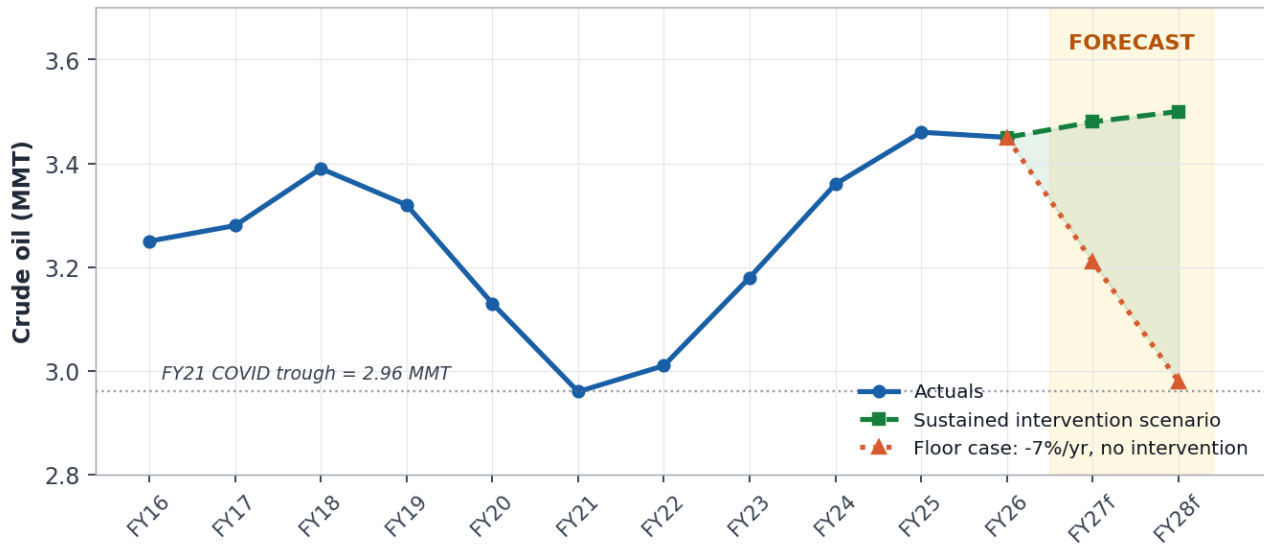
Source: 10_Years_Production_and_Reserves_Data.xlsx, Sheet1: 2P oil column F (rows 3–12); 2P recoverable gas column H (rows 3–12).

Insight 2. RRR has remained below 1.0 for three consecutive years. Across the decade, 2P oil reserves declined 14% while 2P gas rose 15% — a structural gasification of the reserve base. Derived Reserve Life Index stands at approximately 20 years for oil and 43 years for gas at current production rates.

Assumption: Reserve Life Index is derived as 2P reserves divided by FY25 production: $69.39 \text{ MMT} \div 3.46 \text{ MMT} = 20.1$ years for oil; $139.03 \text{ BCM gas at } 0.9 \text{ MMTtoE/BCM, divided by gas-equivalent production, yields } \approx 43$ years for gas. SME to confirm whether OIL prefers a three-year rolling production basis or an end-of-period methodology.

3 Production forecast fan (FY27–FY28)

Predictive panel — illustrative. Coefficient fitting is pending; scenario lines reflect stated assumptions only.



Model specification

$$\text{Production}(t) = \beta_0 + \beta_1 \cdot \text{Workovers}(t) + \beta_2 \cdot \text{Workovers}(t-1) + \beta_3 \cdot \text{DevWells}(t) + \beta_4 \cdot \text{DevWells}(t-1) + \beta_5 \cdot \text{ExplWells}(t) + \beta_6 \cdot \text{ExplWells}(t-1) + \varepsilon$$

β_0 represents the Arps decline baseline — what mature fields deliver with no intervention. Coefficients on same-year (t) terms are expected to be smaller than lag ($t-1$) terms because wells spudded mid-year contribute only partial-year production. Workovers act faster than wells, so β_1 typically exceeds β_2 .

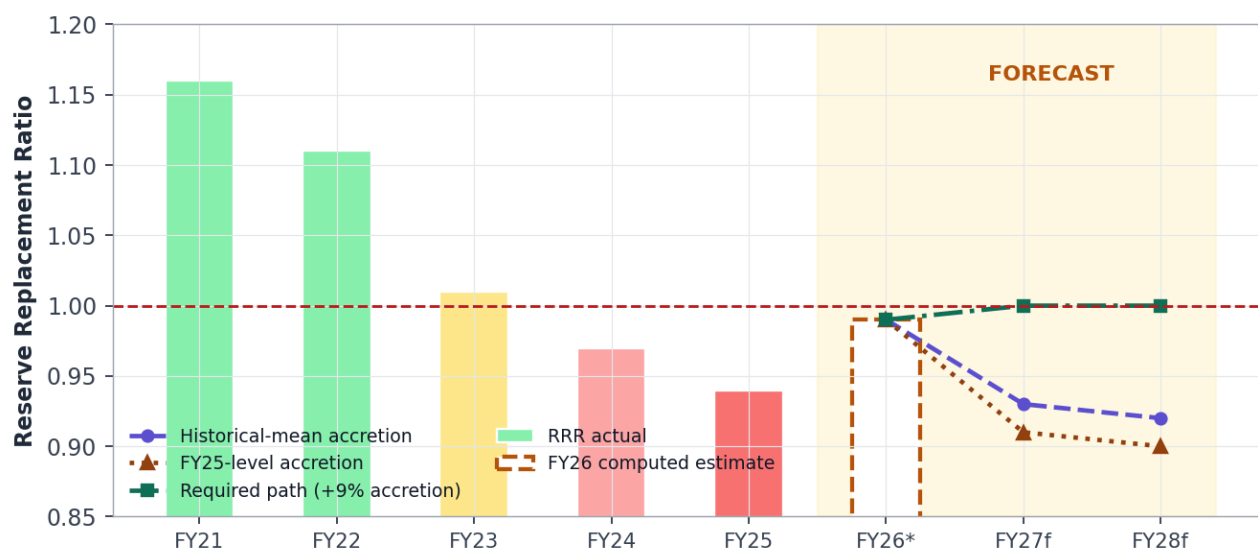
Scenario assumptions

Assumption: Floor case = -7%/yr Arps decline on the FY26 base of 3.45 MMT, with no contribution from workovers, development wells, or exploratory tie-ins. -7%/yr is the upper bound of the textbook decline range for mature onshore fields (5–8%). OIL-specific decline rate will be calibrated against the FY16–FY21 low-intervention history once state-wise production is extracted from Annual Reports. **Sustained case** assumes intervention levels at FY26 plan (≈307 workovers/yr, ≈52 development wells, ≈22 exploratory wells) — illustrative anchor only.

Under the floor case, FY28 lands at 2.98 MMT — below the FY21 COVID trough of 2.96 MMT. The shaded band between sustained and floor lines quantifies what intervention buys: approximately 0.52 MMT in FY28.

4 RRR scenario fan (FY26–FY28)

Predictive panel — illustrative. Required uplift derived from the arithmetic verification shown below.



Arithmetic verification

Year	Accretion (MMToE)	RRR given	Implied depletion	Production check
FY21	6.03	1.16	5.20	5.34
FY22	6.15	1.11	5.54	5.75
FY23	5.95	1.01	5.89	6.04
FY24	5.85	0.97	6.03	6.22
FY25	5.87	0.94	6.24	6.39

Assumption: Formula assumed: $RRR = \text{annual accretion} \div \text{depletion}$, where depletion is approximated by production in MMToE using $1 \text{ BCM} = 0.9 \text{ MMToE}$. The "implied depletion" and "production check" columns agree within approximately 5%; the residual likely reflects a different gas-conversion factor or LPG/condensate inclusion. SME to confirm OIL's exact RRR formula and gas-equivalent conversion factor.

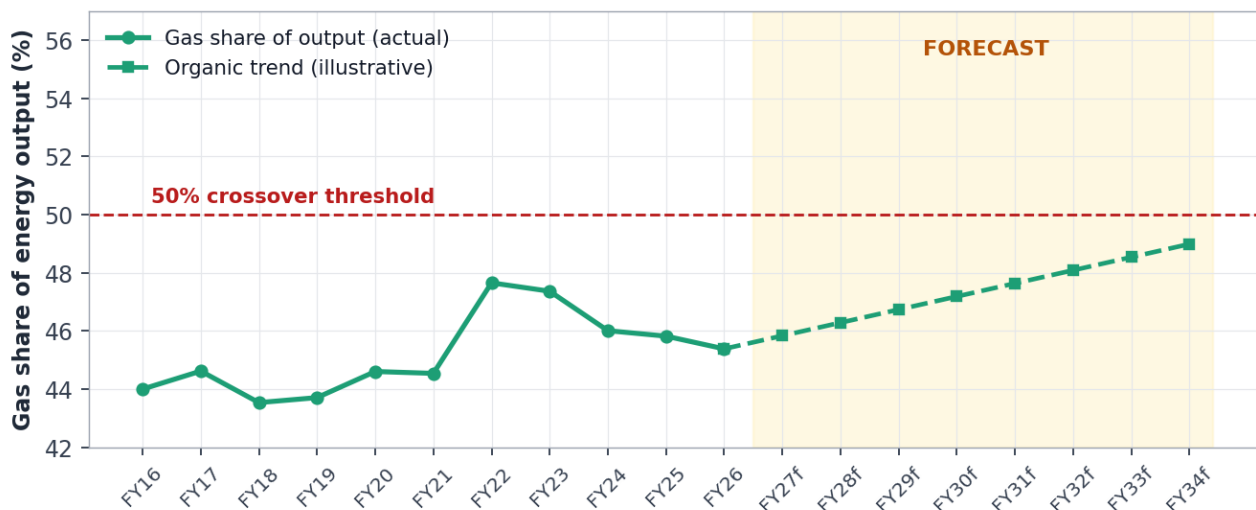
Required uplift derivation

Historical-mean accretion (FY21–FY25) = $(6.03 + 6.15 + 5.95 + 5.85 + 5.87) \div 5 = \mathbf{5.97 \text{ MMToE per year}}$. Required FY28 accretion to hold RRR at 1.0 equals projected FY28 depletion ($\approx 6.50 \text{ MMToE}$). Required uplift = $6.50 \div 5.97 - 1 = \mathbf{+8.9\%}$, rounded to +9%.

FY26 RRR computed estimate ≈ 0.99 (hollow bar): accretion $6.23 \div$ implied depletion $6.32 = 0.986$. Marked "FY26*" pending publication of the FY26 Annual Report for confirmation.

5 Gasification crossover projection

Predictive panel — illustrative. Organic drift modelled at +0.45 pp/yr.



Claim. Natural gas is already approximately 45% of company output on an energy-equivalent basis. Under organic-drift assumptions, the 50% crossover occurs around FY32–FY34. A natural next layer is joining quarterly realisation data to convert the volume shift into a revenue-mix forecast. The Andaman frontier scenario is treated separately on page 8.

Assumption: Gas share computed using $1 \text{ BCM} \approx 0.9 \text{ MMToE}$. Organic drift rate of +0.45 percentage points per year is the historical FY16–FY26 average annual movement, used as an illustrative anchor — not a fitted forecast model.

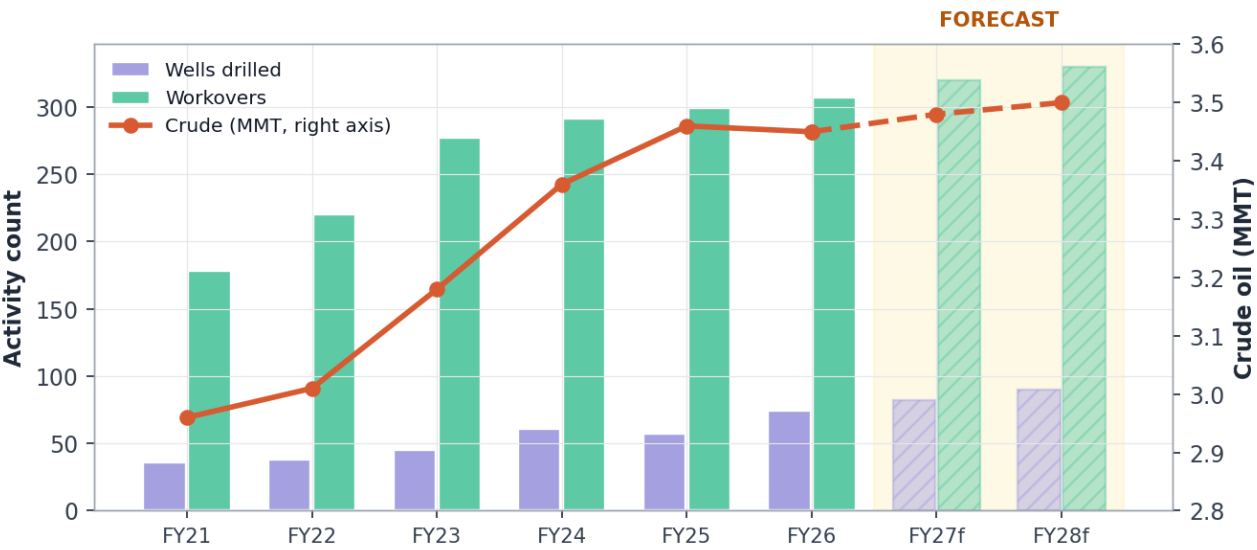
6 Exploration page — KPIs and analytics

WELLS DRILLED 74 FY26 · Nominated 67 + Other 7	DRILLING METERAGE 92.9% FY26 · 246.3k of 265k m	WORKOVERS 307 FY26 · vs 299 in FY25	RESERVE ACCRETION 6.23 MMTToE FY26 · +6.1% YoY · best in 5 yrs
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Source: Wells and meterage from FY202526_Performance.xlsx: Annexure-III-Expl.Drl row 61 (exploratory: 22 wells, 97,421 m), Annexure-IV-Dev.Drl row 14 (development: 52 wells, 148,915 m), row 15 (grand total: 74 wells, 246,336 m of 265,038 m target). Workovers from Workover__Drilling_5_yrs.xlsx row 5. Accretion from 10_Years_Production_and_Reserves_Data.xlsx row 13 column L.

FY26 drilling breakdown — nominated vs. other regimes

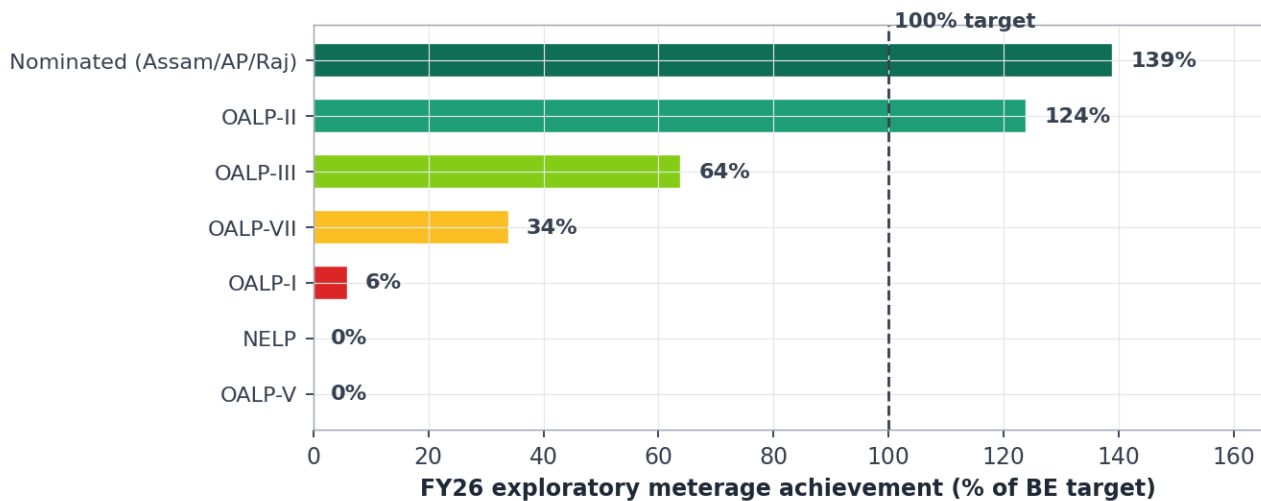
Category	Nominated	Other	Total
Exploratory wells	17	5	22
Development wells	50	2	52
Total wells	67	7	74
Total meterage (m)	223,063	23,273	246,336



Source: Wells FY21–FY25 from Workover__Drilling_5_yrs.xlsx rows 4–8; FY26 as cited above. Workovers FY21–FY26 from Workover__Drilling_5_yrs.xlsx row 5 (Total). Crude FY21–FY26 from 10_Years_Production_and_Reserves_Data.xlsx. FY27–FY28 hatched bars are illustrative forecast extensions.

6 Exploration page — regime achievement detail

The FY26 meterage achievement breaks down materially by licensing regime. Nominated blocks over-delivered; OALP-II — driven by the Andaman East campaign — was the second-strongest performer. NELP, OALP-V, OALP-I and OALP-VII were the principal drags.

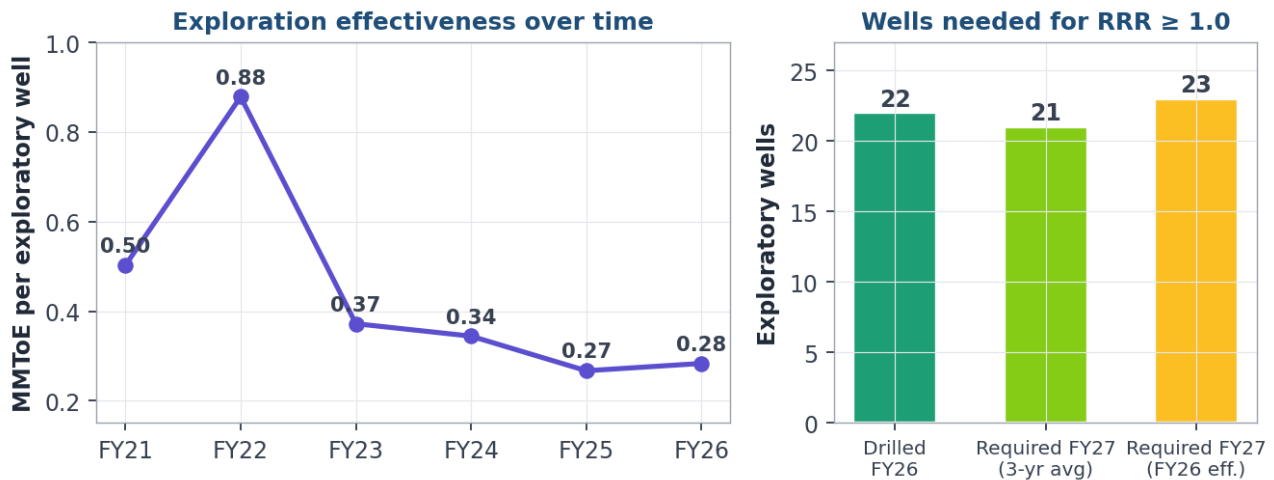


Source: FY202526_Performance.xlsx, sheet Annexure-III-Expl.Drl, subtotal rows: row 11 Nominated 139%, row 16 NELP 0%, row 26 OALP-I 6%, row 33 OALP-II 124%, row 40 OALP-III 64%, row 45 OALP-V 0%, row 51 OALP-VII 34%. Achievement values compare cumulative meterage drilled through March 2026 against annual BE targets.

Insight 3. Execution remains concentrated in legacy nominated acreage. The OALP-II performance shows frontier blocks can deliver when prioritised; conversely, NELP and OALP-V at zero indicate commitment-risk exposure that compounds across years. OALP-II's overdelivery is largely attributable to the Andaman East well (block AN-OSHP-2018/1, 8,456 m drilled — 199% of target). The Andaman result is treated as a separate possible-upside scenario on page 8.

7 Exploration effectiveness and required-wells inversion

Predictive panel — illustrative. Effectiveness ratio derived per exploratory well.



Source: Exploratory well counts from Workover__Drilling_5_yrs.xlsx (FY21–FY25) and FY202526_Performance.xlsx Annexure-III row 61 (FY26: 22 wells). Reserve accretion from 10-year Excel column L. FY26 effectiveness = 6.23 MMTcE ÷ 22 wells = 0.28 MMTcE/well.

Claim. Accretion per exploratory well has fallen from 0.50 MMTcE (FY21) to 0.27 MMTcE (FY25), recovering slightly to 0.28 in FY26 — more wells are buying less reserve. Inverting at current effectiveness, FY27 requires 21–23 exploratory wells to restore RRR ≥ 1.0, against 22 actually drilled in FY26. This panel cross-references the RRR scenario fan on page 4: the two pages solve the same equation from opposite ends.

Assumption: Required FY27 accretion is assumed equal to projected FY27 depletion of approximately 6.42 MMTcE (linear extrapolation of the depletion trend from page 4). Effectiveness assumed constant at the recent average. SME to validate whether per-well effectiveness is the right denominator versus per-metre or per-rupee-capex, and whether the accretion numerator should net out revisions before computing effectiveness.

8 Andaman scenario module — possible upside

Scenario — illustrative · unbooked. Excluded from all 2P, accretion and RRR computations. No official reserve estimate has been published; analog volumes below are illustrations, not estimates.

Source citations for Andaman content

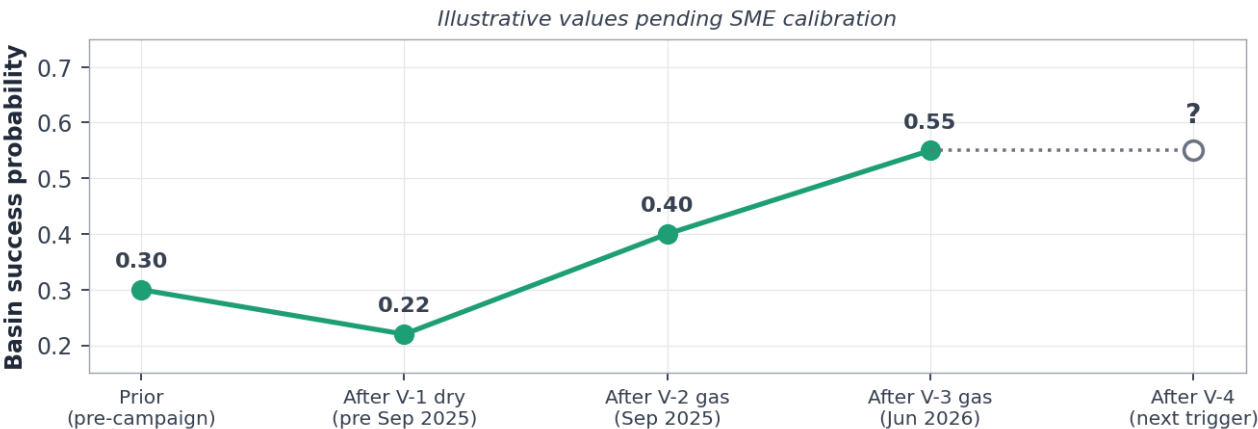
Item	Reference
Vijayapuram-3 primary disclosure	Oil India Limited stock-exchange disclosure under SEBI LODR Regulation 30, 5 June 2026 — "Oil India reports natural gas in Vijayapuram-3 well in Andaman block; appraisal drilling planned." Available at bseindia.com (corporate-announcement filings for OIL India, 5 June 2026) and nseindia.com.
Vijayapuram-2 prior discovery	Announced 26 September 2025. Coverage at newsonair.gov.in; outlookbusiness.com ("Oil India Discovers Natural Gas in Andaman Shallow Offshore Block").
Vijayapuram-1 well status	Referenced in OIL India Q2 FY26 earnings-call transcript filed on BSE (21 November 2025). Drilled prior to Vijayapuram-2; not a gas success. Associated exploratory write-off of approximately Rs 720 crore reported in analyst-research coverage of the Q2 FY26 result.
Vijayapuram-3 corroborating coverage	Press Trust of India via siasat.com; wionews.com; businessstoday.in (5 June 2026 reports).
Block identifier in internal MIS	AN-OSHP-2018/1 (Andaman East) — matches FY202526_Performance.xlsx, Annexure-III row 27 (8,456 m drilled in FY26, 199% of meterage target).

Campaign well status

Well	Status	Date	Location
Vijayapuram-1	Dry — write-off ≈ Rs 720 crore	Pre-Sep 2025	Andaman offshore
Vijayapuram-2	Gas — 87% methane	Sep 2025	17 km offshore · 295 m water depth
Vijayapuram-3	Gas — continuous flaring	Jun 2026	15 km offshore · 355 m water depth

Bayesian basin-probability tracker

A prior on basin success is updated with each well result: V-1 (dry) lowered the prior; V-2 and V-3 (gas) raised it. Values shown are illustrative for SME calibration; the next update trigger is the V-4 spud.



8 Andaman scenario module — hypothetical sensitivities

Hypothetical bookings — illustrative analogs only. No public reserve estimate exists for the Vijayapuram cluster. The brackets below model the impact *if* a reserve of the stated size were ultimately booked.

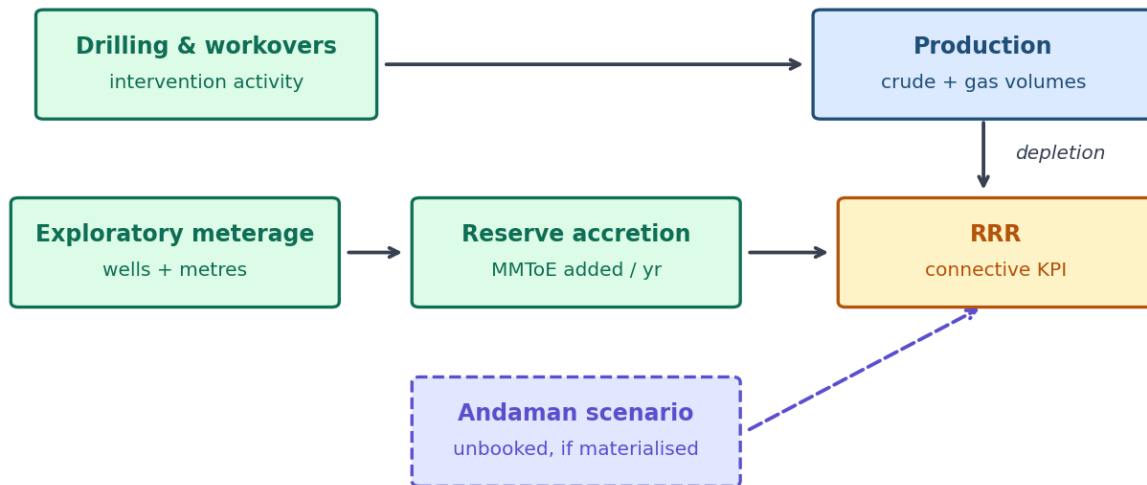
	25 BCM (small)	50 BCM (mid)	75 BCM (large)
Uplift to current 2P gas (139 BCM)	+18%	+36%	+54%
Reserve life added (at FY25 gas prod.)	≈ +8 years	≈ +15 years	≈ +23 years

Assumption: The 25 / 50 / 75 BCM brackets are illustrative analog volumes, not OIL or DGH estimates. Reserve-life math: incremental BCM × 0.9 MMT_{OE}/BCM ÷ FY25 gas-equivalent production (≈ 2.93 MMT_{OE}). SME to replace these brackets with geologically comparable analog fields, or to strike this sensitivity panel entirely if any quantitative anchor is considered premature.

What materialisation would mean

Frontier-to-production timelines for deepwater offshore typically run 7–10 years from discovery confirmation to first commercial gas. Any Andaman contribution to OIL's reserve base and gasification trajectory therefore lands from approximately FY32 onwards. Until reserves are booked, the dashboard's RRR, 2P and accretion KPIs exclude Andaman by design (record-level *booked* = *false* flag).

9 Cross-page predictive logic map



RRR is the connective KPI across pages. It appears on the production page as an outcome, on the exploration page as the target the accretion model solves for, and in the Andaman module as the metric the scenario could reset. Solid arrows are fitted or directly-derived relationships. The dashed arrow from the Andaman scenario indicates a non-binding overlay — it never enters KPI math.

Demonstration of analytical depth. A single click on RRR from any of the three entry points opens the same drill-down, making the dashboard's cross-domain reasoning tangible. This is the architectural pattern recommended for productionisation: shared KPIs as graph nodes, page-specific lenses as views.

10 Predictive claims inventory

Every predictive statement in this pack falls into one of three tiers. Lead with fitted models that have been validated against the FY26 known year; present scenario simulations with assumptions visible; never present derived arithmetic as machine learning.

#	Predictive claim	Tier	Shown in
1	FY27–28 production forecast (decline + intervention regression)	Fitted model — pending	\$3
2	RRR trajectory — accretion +9% needed by FY28	Scenario simulation	\$4
3	Reserve Life Index — oil $\approx$ 20 yrs, gas $\approx$ 43 yrs	Derived arithmetic	\$2
4	Gas share of output crossing 50% (organic trend)	Scenario simulation	\$5
5	Intervention ROI — tonnes per workover / dev well / expl well	Fitted model — pending	\$3, \$6
6	NELP / OALP commitment-risk scoring	Scenario simulation	\$6
7	Exploration effectiveness $\rightarrow$ wells needed for $RRR \geq 1.0$	Fitted model — pending	\$7
8	Bayesian basin probability, updated per Andaman well	Scenario simulation	\$8
9	Hypothetical Andaman booking sensitivities	Scenario simulation	\$8
10	First gas approximately FY32+ (frontier timeline norm)	Derived arithmetic	\$8

The "scenario simulation" tier is the most assumption-heavy. Each scenario panel in this pack declares its formula and constants in an Assumption box adjacent to the chart so that the reviewer can challenge each assumption independently.

A1 AI extraction discipline — failure modes to prevent

This appendix is for the development team. It documents the classes of error an AI extraction and analytics agent can introduce when working with the OIL data corpus, and the corrective principle that prevents each. Cross-references in Appendix A2 (Prompt P-n) map each principle to a paste-ready prompt snippet.

F1 Inference presented as direct readout

What this looks like	A claim describes a multi-year trend (for example "Assam base decline") when the source file only contains a single-year target-versus-actual snapshot.
Corrective principle	Every statement that goes beyond what a file row literally contains must be tagged as inference, with the underlying assumption shown alongside. See P2.

F2 Arbitrary modelled rate with no declared basis

What this looks like	A scenario line is plotted using a rate (decline, growth, conversion) that does not appear anywhere in the source data and is not declared on the chart.
Corrective principle	Every modelled or projected line must declare its rate, formula and source on the chart itself, not buried in text. See P6.

F3 Round-number placeholder treated as analytical insight

What this looks like	A statement such as "X must rise 25%" is emitted where the actual arithmetic from the data yields a different figure (for example 8.9%).
Corrective principle	Quantitative claims must be derived from the data and shown with the calculation. Round only at the final display step. See P7.

F4 Year mismatch across one KPI panel

What this looks like	A dashboard strip mixes KPIs from different fiscal years (for example wells-drilled as FY25 next to meterage-achievement as FY26).
Corrective principle	All KPIs on a single panel must use the same fiscal year. Backfill older-year KPIs from newer sources if available. See P5.

F5 Approximation when exact data exists

What this looks like	A figure is emitted as approximate ("≈72 wells") when the source file contains an exact grand-total row (74 wells).
Corrective principle	Approximation language is prohibited in front of any value that can be read exactly from a file. See P3.

F6 Category lumping that hides the truth

What this looks like	Sub-categories with very different values are aggregated into a parent group — for example, lumping NELP (0%) with OALP-II (124%) under a single label.
Corrective principle	When a source file provides subtotal rows, use them. Lump only when the values are homogeneous; otherwise display sub-categories separately. See P4.

F7 External claim with no citation

What this looks like	A factual claim is made without naming its source, leaving the reader unable to verify or audit.
Corrective principle	Every datum has a source attached: file + sheet + row, OR a URL with publisher, title and date. See P1.

F8 Text-chart mismatch (predictions invisible)

What this looks like	Text mentions FY27–FY28 forecasts but the corresponding chart only plots actuals up to the current year.
Corrective principle	Every predictive claim in the text must have a visually distinct forecast region on the chart, with shading and an explicit label. See P8.

F9 Incomplete data extraction

What this looks like	A multi-year chart stops at FY25 even though FY26 actuals exist in a different file in the same corpus.
Corrective principle	Before plotting any series, scan every file in the corpus for the latest available year. See P10.

F10 Unverified formula or conversion factor

What this looks like	A derived metric is used in a forecast without showing that the formula reproduces the historical series.
Corrective principle	Every derived metric must declare its formula and constants, and pass an arithmetic verification table against at least three historical years. See P11.

F11 Survivorship bias on success metrics

What this looks like	A success-rate statistic counts only successful attempts in the denominator (for example "2 wells confirmed" without mentioning the dry V-1 well).
Corrective principle	Every attempt — including dry wells, write-offs and failed bids — must appear in the denominator of any success-rate statistic. See P12.

F12 Aggregation not validated against grand totals

What this looks like

Component values are summed without checking against the file's own grand-total row, allowing extraction errors to go undetected.

Corrective principle

Whenever a grand-total row exists in the source, the agent must compute the breakdown and verify the components sum to the total. Discrepancies signal an extraction failure. See P3.

A2 AI developer prompts — guardrails for the production system

Drop-in prompt snippets paired with the failure modes from Appendix A1. Each prompt is designed to be installed at one of four architectural layers: system prompt, tool description, output validator, or workflow gate. The mapping is shown at the end of this section.

P1 Source-citation discipline Addresses F1, F7

Every numeric or factual statement in your output MUST include a source attribution in the format:
[source: {file_name}, sheet "{sheet_name}", row {row_num}, column "{column_label}"]
OR for external sources:
[source: {publisher}, "{title}", {date}, URL: {url}, retrieved: {YYYY-MM-DD}]
If a claim cannot be sourced this precisely, do not make the claim. If multiple statements share a source, cite it once per visualisation or insight panel. No floating facts.

P2 Inference vs readout flagging Addresses F1, F2

Classify every statement you generate into one of three categories before emitting:
(a) DIRECT_READOUT – value exists verbatim in a named source row
(b) DERIVED – computed from source values via a stated formula; declare formula and constants
(c) INFERRED – extends beyond source data using a stated assumption; declare the assumption
Render (b) and (c) statements with a visible flag in the user-facing output.
Refuse to combine categories silently.

P3 No approximations when exact data exists Addresses F5, F9, F12

When reading from a file, you MUST scan every relevant row and use the exact value.
The strings "~", "approximately", "about", "around" are FORBIDDEN in front of any number unless that number is a forecast/scenario projection.
For totals: if a file contains an explicit grand-total row, use that row AND verify your component breakdown sums to it. Report any sum mismatch as a data-quality alert.

P4 Category-integrity check Addresses F6

Before aggregating sub-categories into a parent group, compute per-category min, max and range.
If max/min > 5, or if the range spans both <10% and >100%, the categories are TOO HETEROGENEOUS to lump – display them separately.
Always prefer the source file's own subtotal rows over ad-hoc groupings.
When in doubt, show both the subtotal and the disaggregated bars.

P5 Year-alignment validator Addresses F4

For every multi-KPI dashboard panel: extract the fiscal_year of each KPI before rendering.
If any two KPIs on the same panel use different fiscal_years, raise an alignment warning and use the newest year available in any source file as the canonical year.
Backfill older-year KPIs from later sources if available.
Never ship a panel with mixed years unless explicitly comparative.

P6 Assumption-declaration on charts Addresses F2, F10

Any chart containing a modelled, forecast, or derived series MUST render an Assumptions caption beneath it listing:

- (a) the formula or model specification
 - (b) every numeric constant used (rates, conversion factors, lags)
 - (c) the data window the model was fitted on
 - (d) the date the calculation was last refreshed
- A chart without this caption is incomplete and must not be released.

P7 Round-number suspicion Addresses F3

When generating quantitative claims (e.g. "X must rise by N%"), suspect any value that is suspiciously round: 5, 10, 15, 20, 25, 50, 100.

Before emitting such a value, recompute it from the underlying data and replace with the actual arithmetic result (e.g. +8.9% not +9%).

Round only at the final display step, and show the unrounded value in a tooltip or footnote.

P8 Forecast visibility Addresses F8

If your text output contains the words "forecast", "projection", "predict", "scenario", "expected", "if ... then", or any future fiscal_year reference relative to current_year, the accompanying chart MUST include a visually distinct forecast region: shaded background, dashed lines beyond actuals, and a "FORECAST" label.

A predictive claim with no visible forecast in the chart is a contract violation.

P9 External-vs-internal source segregation Addresses F7, F11

Tag every data record at ingestion with:

origin = INTERNAL_FILE | EXTERNAL_DISCLOSURE | EXTERNAL_PRESS | EXTERNAL_ANALYST

External records get an additional booked: false flag if they describe unbooked discoveries, scenarios, or speculative figures.

Aggregation functions (sum, mean, projection) must filter origin = INTERNAL_FILE

AND booked = true by default.

External data may inform scenarios but never enter KPI math.

P10 Full-history extraction Addresses F9

For every metric request: scan ALL files in the corpus, identify the latest fiscal_year present for that metric, and extract every year between earliest and latest.

Do not stop at the first file or accept a stale series.

If metric X appears in file A through FY25 and file B has FY26, the agent must use file B for FY26 and report which file each year came from.

P11 Formula-verification step Addresses F3, F10

For any DERIVED metric (RRR, intensity, ratio): before using it in a forecast, run a verification table comparing the derived value to a closed-form computation for at least three historical years.

If the verification gap exceeds 10%, halt and report the methodology mismatch rather than proceeding with a model built on an unverified formula.

P12 Survivorship-bias guard Addresses F11

When reporting outcome statistics (success rates, hit ratios, win rates), include all attempts in the denominator, not just successes.
For exploration campaigns specifically: every well drilled, including dry / written-off, must appear in well-status tables.
Include exploratory write-off events from financial filings as evidence of unsuccessful wells when no positive disclosure exists.

P13 SME-calibration checkpoint Addresses all scenario panels

For any output containing forecast scenarios, model coefficients, or analog volumes: mark with status = AWAITING_SME_CALIBRATION and display placeholder values with "illustrative" badging.
Provide a calibration form: assumption name | placeholder value | SME-confirmed value | rationale.
Do NOT propagate placeholder values into downstream KPI calculations.

Installation by architectural layer

Layer	Prompts to install	Why
System prompt	P1, P2, P3, P7, P8	Universal disciplines applying to every output.
Tool description	P4, P5, P10, P11, P12	Embedded inside data-extraction and aggregation tool descriptions, surfaced when the agent picks the tool.
Output validator	P6, P8, P9	Deterministic post-processing checks on the agent's response; violations are returned to the agent for self-correction.
Workflow gate	P13	Human-in-the-loop step before any illustrative number is allowed to propagate.